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B.Tech. Degree VII Semester Examination November 2016

ME 1701 REFRIGERATION AND AIR CONDITIONING

(2012 Scheme)

Time : 3 Hours

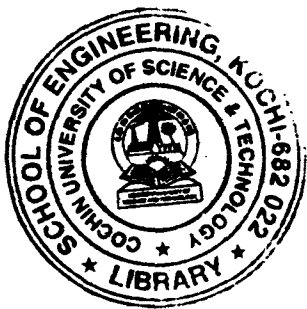
Maximum Marks : 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Define the terms 'refrigeration capacity' and 'coefficient of performance' in relation to a refrigeration system. Why is the term 'efficiency' not used to indicate the performance in the case of refrigerators and heat pumps?
- (b) Discuss the limitations of the Reversed Carnot cycle with gas as refrigerant.
- (c) Describe the actual vapor compression refrigeration cycle on T-s and p-h diagrams.
- (d) Enumerate all the desired properties of refrigerants.
- (e) Differentiate between centrifugal compressor and reciprocating compressor from the point of view of effect of evaporator temperature and condenser temperature on their cooling capacity and power requirement.
- (f) Describe the relative advantages and disadvantages of air-cooled and water cooled condensers.
- (g) Define relative humidity, specific humidity and dew-point temperature and describe theoretical method for determining their values.
- (h) Define the term 'Effective Temperature' and describe the factors which affect Effective Temperature.



PART B

(4 × 15 = 60)

- II. In a Bell-Coleman refrigeration system the outlet temperature from the cold space is 270 K and the temperature at inlet to turbine is 318 K. The pressure ratio is 4.0. Determine the mass flow rate, heat rejection, compressor work, turbine work, COP and the volume flow rates at inlet to compressor and outlet of turbine for a system of 1 TR cooling capacity. The working substance is air, $c_p = 1.005 \text{ kJ/kg K}$, $R = 0.287 \text{ kJ/kg K}$. (15)
- OR**
- III. A simple air cooled system is used for an aero plane having a load of 10 TR. The atmospheric pressure and temperature are 0.9 bar and 283 K respectively. During ramming, pressure increases to 1.013 bar. In the heat exchanger, the temperature of air is reduced by 55°C. The pressure in the cabin is 1.01 bar and the temperature of air leaving the cabin is 298 K. Determine (i) power required to take the cooling load in the cabin (ii) COP of the system. (15)

(P.T.O.)

- IV. A simple saturation cycle using R-12 is designed for taking a load of 10 TR. The refrigerator and ambient temperatures are 273 K and 303 K respectively. A minimum temperature difference of 5°C is required in evaporator and condenser for heat transfer. Find (i) mass flow rate through the system (ii) power required (iii) cylinder dimensions assuming $L/D = 1.2$, for a single cylinder, single acting compressor if it runs at 300 rpm with volumetric efficiency = 0.9. (15)

OR

- V. Explain with a neat sketch the working of a water-lithium bromide absorption refrigeration system. What are its limitations? (15)

- VI. (a) Describe the working principle of Thermostatic Expansion Valve (TEV) with the help of a neat sketch. (8)
- (b) Explain the constructional features of flooded type evaporator. (7)

OR

- VII. An R-12 two cylinder reciprocating compressor is designed for 7.5 tons refrigeration capacity, at 5°C evaporating temperature and 40°C condensing temperature. The clearance is 5% of the stroke and piston speed is limited to 3 m/s. Take $L/D = 0.8$ and compression follows the law $p v^{1.5} = C$. Pressure drops, at suction and discharge valves may be taken as 0.2 bar and 0.4 bar respectively. Determine the following: (i) power required to run the compressor and COP (ii) volumetric efficiency of the compressor (iii) r.p.m. of the compressor and its dimensions. (15)

- VIII. (a) Explain various psychrometric processes. (7)
- (b) In a cooling application, moist air enters a refrigeration coil at the rate of 100 kg/min at 35°C and 50% RH. The apparatus dew point of coil is 5°C and bypass factor is 0.15. Determine (i) outlet state of moist air (ii) cooling capacity of coil. (8)

OR

- IX. (a) Explain summer and winter air conditioning. (8)
- (b) 250 m³/min of air at atmospheric conditions 12°C DBT and 50% Relative Humidity is supplied to an air-conditioned hall. The required conditions are 18°C DBT and 60% Relative Humidity. Determine (i) sensible heat and latent heat removed from the air per minute and (ii) sensible heat factor for the system. (7)